

3206 67th St W, Bradenton,
Florida, 34209

Ben-Larbi

GENERAL NOTES

1. DESIGN CRITERIA

FLORIDA BUILDING CODE (FBC), 8TH EDITION (2022),
ACI 318-19, BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE,
ASTM STANDARDS AS SPECIFIED HEREIN,
ALUMINUM DESIGN MANUAL, 2020 EDITION (AA-ADM-2(20)),
AA-ASM 35-36, ALUMINUM STANDARDS AND DATA,
NATIONAL ELECTRICAL CODE (NEC), 2023 EDITION.

2. DESIGN LOADS

WIND SPEED (PER CHAPTER 16): 140 MPH
ULTIMATE WIND SPEED (VULT) = 138 MPH (PER ASCE 7-22, SECTION 2.6.5.1).
ALLOWABLE STRESS DESIGN WIND SPEED (VASD) = 107 MPH (VASD = VULT × 0.6).
RISK CATEGORY: 1 (LOW HAZARD TO HUMAN LIFE).
EXPOSURE CATEGORY: B (URBAN OR SUBURBAN TERRAIN WITH OBSTRUCTIONS).
DESIGN WIND PRESSURES INTERPOLATED FROM ASCE 7-22 AND REFERENCE TABLES FOR VULT = 140 MPH, EXPOSURE CATEGORY B.

WIND LOADS (MMFRS), BASED ON 140 MPH WIND SPEED AND EXPOSURE B

- HORIZONTAL PRESSURE (WINDWARD): 26 PSF.
- HORIZONTAL PRESSURE (LEEWARD): 21 PSF.
- VERTICAL PRESSURE ON SCREEN SURFACE: 8 PSF
- VERTICAL PRESSURE ON SOLID SURFACES: 27 PSF

LOAD REDUCTION FACTORS:

- FOR ALLOWABLE STRESS DESIGN PRESSURES: 0.6 (ASCE 7-22 SECTION 2.4).
- FOR 20x20x.011" MESH SCREEN: 0.88 (MANUFACTURER DATA, ASCE 7-22 SECTION 30.5)

3. CONTRACTOR RESPONSIBILITIES

ALL WORK SHALL BE PERFORMED BY A LICENSED CONTRACTOR IN ACCORDANCE WITH APPLICABLE CODES, REGULATIONS, AND LOCAL ORDINANCES. PERMITS AND APPROVALS FROM ALL NECESSARY AGENCIES SHALL BE OBTAINED PRIOR TO COMMENCEMENT OF CONSTRUCTION. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS AND APPROVALS. ANY DISCREPANCIES, ERRORS, OR OMISSIONS SHALL BE REPORTED TO THE ENGINEER IN WRITING WITHIN 10 BUSINESS DAYS OF RECEIPT AND PRIOR TO CONSTRUCTION. THE CONTRACTOR SHALL PROVIDE ALL REQUIRED TEMPORARY SHORING, BRACING, AND SAFETY MEASURES DURING CONSTRUCTION. THE CONTRACTOR SHALL PROTECT EXISTING STRUCTURAL UTILITIES, AND LANDSCAPING INCLUDING THE SCOPE OF THIS PROJECT. INSTALLATION PRACTICES MUST COMPLY WITH MANUFACTURER RECOMMENDATIONS, INCLUDING FLASHING AND WATERPROOFING DETAILS. ALL UNDERGROUND UTILITIES SHALL BE LOCATED AND PROTECTED PRIOR TO EXCAVATION. AS-BUILT DRAWINGS AND MATERIAL CERTIFICATIONS SHALL BE PROVIDED UPON REQUEST. FASTENER TORQUE AND EMBEDMENT TESTING RESULTS SHALL BE PROVIDED IF REQUESTED.

4. MATERIALS

FOUNDATION

- ASSUMED SOIL BEARING CAPACITY: 2,000 PSF (VERTICAL FIELD)
- NOTIFY ENGINEER IF UNSUITABLE SOILS ARE ENCOUNTERED.
- APPLY TERMITRE TREATMENT AS REQUIRED BY FPC.
- PLACE CONCRETE OVER VAPOR BARRIER WHERE APPLICABLE.

CONCRETE

- MINIMUM COMPRESSIVE STRENGTH: 3,000 PSI @ 28 DAYS.
- PORTLAND CEMENT: ASTM C150, TYPE I.

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STEEL REINFORCEMENT

- REINFORCING STEEL: ASTM A615, GRADE 60.
- MINIMUM COVER FOR FOOTINGS: 2 INCHES.
- LAP SPICES: 40 TIMES BAR DIAMETER.
- WELDED WIRE MESH PER ASTM A185.
- FIBERMESH MAY BE SUBSTITUTED IF ASTM AND ACI REQUIREMENTS ARE MET.

ALUMINUM

- STRUCTURAL ALUMINUM ALLOY: 6005-T5 MINIMUM.
- DESIGN PER AA ADM-2020 SPECIFICATIONS.
- MINIMUM WALL THICKNESS FOR EXTRUDED MEMBERS: 0.040 INCHES.
- SELF-MATING BEAMS TO BE STITCHED WITH #12 SMS SCREWS;

- 18" O.C. ALONG SPANS.

- 18X14X0.013" MESH SCREEN IS STANDARD UNLESS OTHERWISE APPROVED

- ALL ALUMINUM IN CONTACT WITH STEEL OR PRESSURE-TREATED LUMBER SHALL HAVE DIELECTRIC ISOLATION

- METAL STRUCTURES WITHIN 5 FEET OF POOLS SHALL BE BONDED PER NEC 680.26.

FASTENERS

ALL PASTERNS SHALL COMPLY WITH ASTM A153 FOR ZINC COATING.
-CONCRETE SCREENS, SHAPED, STRENGTHEN, REINFORCED OR APPROVED EQUAL.
-STEEL SCREENS SHALL BE 1/4" THICK, 1/2" SPACING, 1/2" OVERLAP, 1/2" STAPLES.
-STAINLESS STEEL, TYPE 304 (16-8) SHALL BE USED FOR CONNECTIONS TO PRESSURE-TREATED
-MINIMUM PASTER SPACING:
-SMS 3/4" CENTER-TO-CENTER, 1/2" EDGE DISTANCE
-1/2" MEMBERS TO WOOD, #14 X 2-3/4" WOOD SCREWS @ 18" O.C.
-1/2" MEMBERS TO ALUMINUM #10 X 1-1/2" SMS OR TEK SCREWS @ 18" O.C.
-FOR OPEN VIEW OR PICTURE WINDOW ASSEMBLIES, REDUCE PASTER SPACING TO 12" O.C.

5. ELECTRICAL

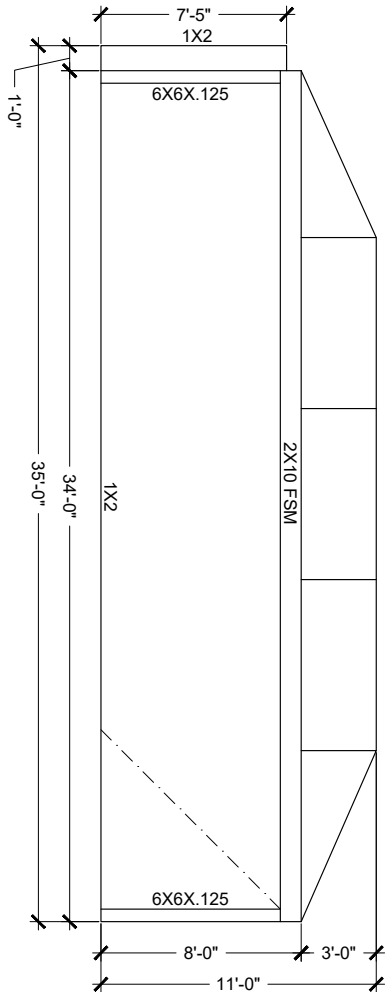
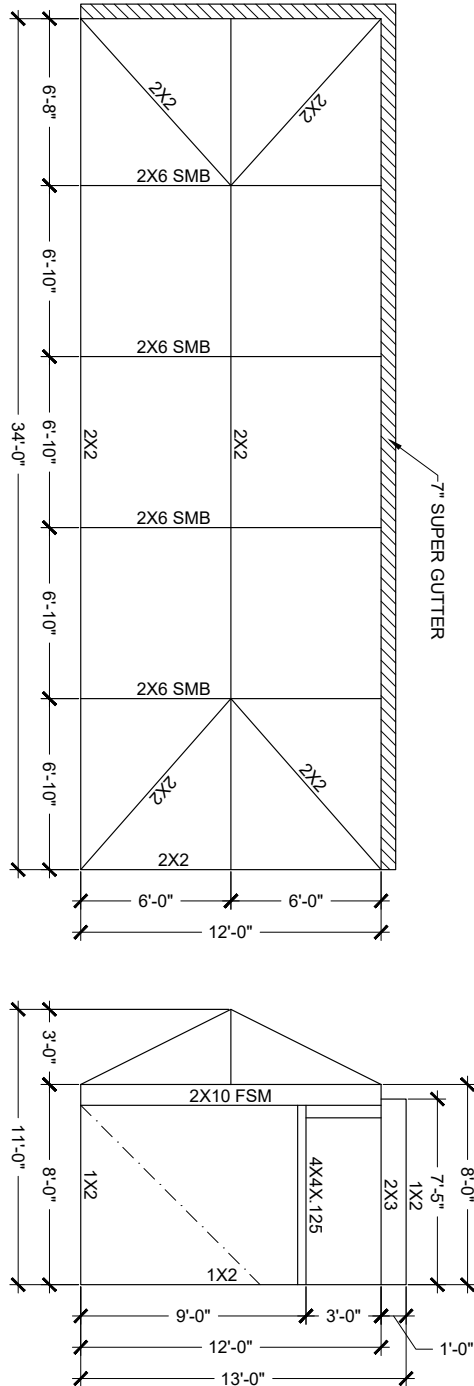
- METAL STRUCTURES WITHIN 5 FEET OF SWIMMING POOLS MUST BE BONDED IN ACCORDANCE WITH NEC 680.22
- PROVIDE EQUIPOTENTIAL BONDING FOR ALL CONDUCTIVE ELEMENTS WITHIN POOL AREA.
- ALL ELECTRICAL WORK SHALL COMPLY WITH NEC 2023 EDITION AND LOCAL AMENDMENTS.

5. GENERAL REQUIREMENTS

- DIMENSIONS ARE NOMINAL UNLESS OTHERWISE NOTED.
- DIMENSIONAL TOLERANCES $\pm .18"$ UNLESS SPECIFIED.
- NOTES APPLY TO THE ENTIRE DRAWING SET UNLESS OTHERWISE INDICATED.
- REPORT ANY CONFLICTS IN DRAWINGS TO THE ENGINEER IMMEDIATELY.
- PROVIDE SHOP DRAWINGS AND MATERIAL CERTIFICATIONS IF REQUESTED.
- ENSURE SITE ACCESS FOR REQUIRED INSPECTIONS.
- FINAL CLEANUP IS REQUIRED PRIOR TO PROJECT CLOSEOUT.

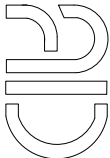
ALUMINUM EXTRUSION DIMENSIONS	
HOLLOW SECTIONS	2 x 2 2" x 2" x 0.060"
	2 x 3 2" x 3" x 0.050"
OPEN-BACK SECTIONS	1 x 2 1" x 2" x 0.044"
SELF-CLEANING SECTIONS (SIB)	
2 x 4 2" x 4" x 0.044" x 0.100"	
2 x 5CL 2" x 5" x 0.050" x 0.135"	
2 x 6 2" x 6" x 0.050" x 0.120"	
2 x 7 2" x 7" x 0.057" x 0.120"	
2 x 8 2" x 8" x 0.072" x 0.124"	
2 x 9 2" x 9" x 0.072" x 0.126"	
2 x 10 2" x 10" x 0.092" x 0.134"	
RECT-TUBE - 60635-T5-MIN.	
	2 x 6 2" x 6" x 0.125" x 0.125"

[illegible]



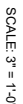
LEGEND:
- - - - - Indicated cable bracing. See details on sheet 3.



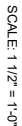
4	CORRECTIONS	JDCV		12/23/2025	NOTICE: THE DATA IN THIS DOCUMENT INCORPORATES PROPRIETARY RIGHTS OF ACTIVE DESIGN LLC. ANY PARTY ACCEPTING THIS DOCUMENT DOES SO IN CONFIDENCE AND AGREES THAT IT SHALL NOT BE DUPLICATED IN WHOLE OR PART NOR DISCLOSED TO OTHERS WITHOUT THE CONSENT OF ACTIVE DESIGN LLC. CONTRACTOR SHALL VERIFY ALL CONDITIONS ON JOB SITE & NOTIFY ENGINEER OF RECORD OF ANY VARIATIONS FROM DIMENSIONS SHOWN IN THESE DRAWINGS BEFORE PROCEEDING WITH ANY CONSTRUCTION.	 Active Design LLC info@activedesigneng.com (310) 963-4514	DATE: 12/9/2025	SHEET: 2
3	CORRECTIONS	JDCV		12/16/2025				
2	CORRECTIONS	JDCV		12/12/2025				
1	INITIAL DESIGN	JDCV	AL	BC				
REV	DESCRIPTION	DRAWN	REVIEWED	APPR.	DATE	PROJECT NAME: Ben-Larbi		



Cable to Frame Connection
SCALE: 3" = 1'-0"



Cable to Foundation Connection

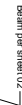


Sloped Purlin to Column Connection



TOP VIEW

1. Fascia to be fastened to rafter tail using (1) 1/4" Ø x 4" lag screw
2. 7" gutter permitted with strap length increased by 2".



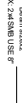
Beam per sheet 0.27

BEAM	QTY-SIZE SMS/SIDE
1	1
2	1
3	1
4	1
5	1
6	1
7	1
8	1
9	1
10	1
11	1
12	1
13	1
14	1
15	1
16	1
17	1
18	1
19	1
20	1
21	1
22	1
23	1
24	1
25	1
26	1
27	1
28	1
29	1
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41	1
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81	1
82	1
83	1
84	1
85	1
86	1
87	1
88	1
89	1
90	1
91	1
92	1
93	1
94	1
95	1
96	1
97	1
98	1
99	1
100	1

Column To Beam Connection



OPTION 1



X: 24 SAB USE 6

NOTE
3/16"-1/5" gusset, fitted with S&B with tight tolerance to prevent slip. Position along top and bottom beam edges. Mainline minimum fastener spacing of 1-1/8" (shop) and 1-1/4" (field). Edge distance: min. 1/2". Slaggap fasteners as needed per adjusted fastener table. Internal puller screws count toward gusset connections. Pattern applies to A-frame configurations.

Gusset Connection

BEAM SIZE	QUANTITY	BEAM FACTOR	CONNECTOR
1/2"	100	1.0	1.0
3/4"	100	1.0	1.0
1"	100	1.0	1.0
1 1/4"	100	1.0	1.0
1 1/2"	100	1.0	1.0
1 3/4"	100	1.0	1.0
2"	100	1.0	1.0
2 1/4"	100	1.0	1.0
2 1/2"	100	1.0	1.0
2 3/4"	100	1.0	1.0
3"	100	1.0	1.0
3 1/4"	100	1.0	1.0
3 1/2"	100	1.0	1.0
3 3/4"	100	1.0	1.0
4"	100	1.0	1.0
4 1/4"	100	1.0	1.0
4 1/2"	100	1.0	1.0
4 3/4"	100	1.0	1.0
5"	100	1.0	1.0
5 1/4"	100	1.0	1.0
5 1/2"	100	1.0	1.0
5 3/4"	100	1.0	1.0
6"	100	1.0	1.0
6 1/4"	100	1.0	1.0
6 1/2"	100	1.0	1.0
6 3/4"	100	1.0	1.0
7"	100	1.0	1.0
7 1/4"	100	1.0	1.0
7 1/2"	100	1.0	1.0
7 3/4"	100	1.0	1.0
8"	100	1.0	1.0
8 1/4"	100	1.0	1.0
8 1/2"	100	1.0	1.0
8 3/4"	100	1.0	1.0
9"	100	1.0	1.0
9 1/4"	100	1.0	1.0
9 1/2"	100	1.0	1.0
9 3/4"	100	1.0	1.0
10"	100	1.0	1.0
10 1/4"	100	1.0	1.0
10 1/2"	100	1.0	1.0
10 3/4"	100	1.0	1.0
11"	100	1.0	1.0
11 1/4"	100	1.0	1.0
11 1/2"	100	1.0	1.0
11 3/4"	100	1.0	1.0
12"	100	1.0	1.0
12 1/4"	100	1.0	1.0
12 1/2"	100	1.0	1.0
12 3/4"	100	1.0	1.0
13"	100	1.0	1.0
13 1/4"	100	1.0	1.0
13 1/2"	100	1.0	1.0
13 3/4"	100	1.0	1.0
14"	100	1.0	1.0
14 1/4"	100	1.0	1.0
14 1/2"	100	1.0	1.0
14 3/4"	100	1.0	1.0
15"	100	1.0	1.0
15 1/4"	100	1.0	1.0
15 1/2"	100	1.0	1.0
15 3/4"	100	1.0	1.0
16"	100	1.0	1.0
16 1/4"	100	1.0	1.0
16 1/2"	100	1.0	1.0
16 3/4"	100	1.0	1.0
17"	100	1.0	1.0
17 1/4"	100	1.0	1.0
17 1/2"	100	1.0	1.0
17 3/4"	100	1.0	1.0
18"	100	1.0	1.0
18 1/4"	100	1.0	1.0
18 1/2"	100	1.0	1.0
18 3/4"	100	1.0	1.0
19"	100	1.0	1.0
19 1/4"	100	1.0	1.0
19 1/2"	100	1.0	1.0
19 3/4"	100	1.0	1.0
20"	100	1.0	1.0
20 1/4"	100	1.0	1.0
20 1/2"	100	1.0	1.0
20 3/4"	100	1.0	1.0
21"	100	1.0	1.0
21 1/4"	100	1.0	1.0
21 1/2"	100	1.0	1.0
21 3/4"	100	1.0	1.0
22"	100	1.0	1.0
22 1/4"	100	1.0	1.0
22 1/2"	100	1.0	1.0
22 3/4"	100	1.0	1.0
23"	100	1.0	1.0

QUANTITY OF #12 SMS PER
TEAM FACE/SIDE/TOTAL FOR
CONNECTION.

QUANTITY OF #14 SMS PER
TEAM FACE/SIDE/TOTAL FOR
CONNECTION.

QUANTITY OF #14 SMS PER
TEAM FACE/SIDE/TOTAL FOR
CONNECTION.

SCALE: 2" = 1'-0"

SCALE: 3" = 1'-0"

SCALE: 3" = 1'-0"

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(310) 963-4514

DATE:	12/9/2025	SHEET:	3
DRAWING TITLE:	DETAILS 1		
PROJECT NAME:	Ben-Larbi		

Ben-Larbi

Provide 1/8" x 1-1/2" x beam height aluminum channel (internal or external). Fasten to column with (8) #14 x 5/8" SMS. Fasten each side with (6) #14 x 5/8" SMS.

EDGE BEAM
PER PLAN

Notch column to receive beam, fasten edge beam to column with (3) 3/8" hex bolts with nut and lock washer.

Provide 2" x 2" x 1/8" aluminum angles, beam height. Fasten each leg with (6) #14 x 5/8" SMS (1" P.C.).

SIDE BEAM
PER PLAN

EDGE BEAM
PER PLAN

Beam to Column Connection
SCALE: 3" = 1'-0"

1/8" x 2" x 2" angle each side of beam (length = beam notch span). Fasten with (4) x 5/8" SMS at 1/2" o.c. #12 x 5/8" SMS at 3/4" o.c.

EDGE BEAM
PER PLAN

0.004" receiving channel, fasten channel to side of beam with (4) #12 x 5/8" SMS (internal). Fasten purlin to channel with (2) #12 x 5/8" SMS each side.

PURLIN
PER PLAN

Beam Under Edge Beam Detail
SCALE: 3" = 1'-0"

Provide 1/8" internal channel, cut to fit with no gap tolerance. Fasten with (8) #14 x 5/8" SMS through channel into beam. Fasten to equal. For wood, use 1/4" x 2-1/2" galvanized lag screws in lieu of masonry anchors.

PER PLAN

Post Base Connection
SCALE: 3" = 1'-0"

1/8" thick internal receiving channel, fasten to beam with (8) #14 SMS through channel into beam.

BRACE
PER PLAN

BEAM
PER PLAN

FASTEN BRACE TO COLUMN AND BEAM WITH 0.004" RECEIVING CHANNEL. FASTEN CHANNEL TO BEAM AND COLUMN WITH (3) #12x5/8" SMS (INTERNAL). FASTEN CHANNEL TO BRACE WITH #12x5/8" SMS (5) PER SIDE.

COLUMN
PER PLAN

K-Brace Detail
SCALE: 2" = 1'-0"

Maintain fastener edge distance of 5/8" and minimum 3/8" separation between fasteners. Install fasteners around the perimeter of the connection first, then complete the remainder in the field.

3/16" 5052-H32 gusset, cut to fit inside beam with no slip tolerance.

BEAM
PER PLAN

BEAM SIZE
PER PLAN

Example screw pattern shown, screw quantity per table below.

NOTE:
1. Beam splice length to be 2x beam height.
2. Locate beam splice at 1/3 points, staggered at each end of beam.
3. Provide minimum 1" camber in beam to reduce self-weight deflection.

BEAM SIZE	FASTENER SIZE	QTY PER FACE/SIDE/TOTAL
2X4 SMB	#12	6 / 12 / 24
2X5 SMB	#12	6 / 12 / 24
2X6 SMB	#12	10 / 20 / 40
2X7 SMB	#12	12 / 24 / 48
2X8 SMB	#14	12 / 24 / 48
2X9 SMB	#14	18 / 36 / 72
2X10 SMB	#14	18 / 36 / 72

Beam Splice Detail
SCALE: 3" = 1'-0"

1. Fascia shall be secured to rafter tail with (1) 1/4" Ø x 4" lag screw.
2. A 7" gutter may be used, provided strap length is increased by 2".

2" x 2" x 1/8" angle each side of beam. Fasten with #12 x 5/8" SMS at 3/4" o.c. each side. Fasten to masonry host with (2) 1/4" T-anchors, providing minimum 2-1/4" embedment into host.

BEAM
PER PLAN

FASCIA

ROOF

Close top of beam with receiver channel, fasten with #12 x 5/8" SMS, minimum (3) each side.

Beam Under Overhang Detail
SCALE: 3" = 1'-0"

(INTERNAL). Fasten each side of beam into channel with (8) #14 SMS, staggered 3/8" separation between fasteners and from edges of members.

BEAM
PER PLAN

CAGE BEAM
PER PLAN

TOP VIEW

CAGE BEAM
SIDE VIEW

BEAM
PER PLAN

Beam to Edge Beam Detail
SCALE: 1-1/2" = 1'-0"

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DATE: 12/9/2025
DRAWING TITLE: DETAILS CONT.
PROJECT NAME: Ben-Larbi

12/9/2025
AL
BC
12/9/2025

1
INITIAL DESIGN
JDCV
AL
BC
12/9/2025

REV
DESCRIPTION
DRAWN
REVIEWED
APPR.
DATE

DATE: 12/9/2025
SHEET: 4